AnalystLab Africa Internship Program

Week 3: SQL & Data Querying

Project 2: Sales Dataset Analysis

Introduction

This project analyzes a sales dataset using SQL to extract business insights from transactional data. The analysis focuses on revenue performance, customer behavior, product performance, and sales trends over time.

SQL techniques applied include filtering, aggregation, grouping, ranking, window functions, and business-oriented analysis. The objective is to demonstrate practical SQL skills for solving real-world business problems and generating actionable insights.

Dataset Overview

The dataset contains 2,823 sales transactions and 25 attributes, including order information, customer details, product information, sales revenue, geographic data, and order status.

Key fields used in this analysis include:

• SALES
• CUSTOMERNAME
• PRODUCTLINE
• PRODUCTCODE
• COUNTRY
• STATUS
• ORDERDATE
• YEAR_ID
• MONTH_ID

These fields were used to evaluate revenue performance, customer purchasing patterns, and product success.

Revenue Analysis

The sales dataset generated a total revenue of **$10,032,628.85** across all recorded transactions.

This indicates strong overall business performance and provides a baseline for evaluating customer behavior, product performance, and market contribution throughout the analysis.

Key Insight

The company generated over **$10 million** in revenue, demonstrating substantial sales activity across multiple products, customers, and geographic regions.

Top Countries by Revenue

This analysis examined revenue contribution across different countries to identify the company's strongest markets.

Key Findings

- The United States generated the highest revenue at **$3,627,982.83**.
- Spain ranked second with **$1,215,686.92** in revenue.
- France generated **$1,110,916.52**, placing third.
- Australia contributed **$630,623.10** in revenue.
- The United Kingdom generated **$478,880.46** in revenue.

Business Insight

The United States is the company's most important market, generating more than three times the revenue of Spain, the second-highest country. This indicates a strong customer base and market presence in the United States.

The top three countries (USA, Spain, and France) contributed a significant share of total revenue, suggesting that business performance is heavily concentrated in a few key markets.

Analyst Observation

The revenue gap between the United States and Spain is approximately **$2.41 million**, highlighting the dominance of the U.S. market within the dataset. Expanding successful sales strategies from these high-performing regions into lower-performing markets could provide opportunities for future growth.

Top Product Lines by Revenue

This analysis evaluated revenue performance across different product lines to identify the company's most profitable product categories.

Key Findings

- Classic Cars generated the highest revenue at **$3,919,615.66**.
- Vintage Cars ranked second with **$1,903,150.84** in revenue.
- Motorcycles generated **$1,166,388.34**.
- Trucks and Buses contributed **$1,127,789.84**.

- Planes generated **$975,003.57** in revenue.
- Trains recorded the lowest revenue at **$226,243.47**.

Business Insight

Classic Cars represent the company's strongest product category, generating nearly **$4 million** in revenue. The substantial gap between Classic Cars and all other product lines suggests that customer demand is highly concentrated in this category.

The business could prioritize inventory management, marketing campaigns, and product expansion within the Classic Cars category to maximize revenue growth.

Analyst Observation

Classic Cars generated more than twice the revenue of Vintage Cars, the second-highest product line. This indicates a clear customer preference and highlights Classic Cars as the primary driver of sales performance within the business.

Monthly Sales Trend Analysis

This analysis examined revenue performance across different months and years to identify trends and seasonal sales patterns.

Key Findings

- Revenue generally increased over the observed period, indicating business growth.
- November consistently recorded the highest revenue across multiple years.
- November 2004 generated the highest monthly revenue at **$1,089,048.01**.
- November 2003 also showed exceptionally strong performance with **$1,029,837.66** in revenue.
- Revenue increased significantly during the final quarter of each year, particularly between October and November.

Business Insight

The sales data reveals a strong seasonal pattern, with revenue peaking toward the end of the year. This suggests increased customer demand during holiday periods and major sales seasons.

Understanding these seasonal trends can help the business improve inventory planning, staffing decisions, marketing campaigns, and promotional strategies during peak sales periods.

Analyst Observation

Revenue nearly doubled between October 2003 (**$568,290.97**) and November 2003 (**$1,029,837.66**). A similar pattern occurred in 2004, reinforcing the importance of year-end sales activity to overall business performance.

Top Customers by Revenue

This analysis identified the customers contributing the highest revenue to the business.

Key Findings

- Euro Shopping Channel generated the highest revenue at **$912,294.11**.
- Mini Gifts Distributors Ltd. ranked second with **$654,858.06** in revenue.
- Australian Collectors, Co. generated **$200,995.41**.
- Muscle Machine Inc generated **$197,736.94**.
- La Rochelle Gifts contributed **$180,124.90** in revenue.

Business Insight

A small number of customers account for a significant portion of total revenue. Euro Shopping Channel alone generated more than **$912,000**, making it the company's most valuable customer.

Maintaining strong relationships with these high-value customers is critical, as losing even one major customer could have a noticeable impact on overall revenue performance.

Analyst Observation

The revenue generated by Euro Shopping Channel is more than four times the revenue generated by Australian Collectors, Co., the third-highest customer. This indicates a concentration of revenue among a few key customers and highlights the importance of customer retention strategies.

Customer Revenue Ranking

This analysis ranked customers according to their total revenue contribution using SQL window functions.

Key Findings

- Euro Shopping Channel ranked **1st**, generating **$912,294.11** in revenue.
- Mini Gifts Distributors Ltd. ranked **2nd** with **$654,858.06**.
- Australian Collectors, Co. ranked **3rd** with **$200,995.41**.
- The top ten customers each generated more than **$150,000** in revenue.
- Boards & Toys Co. recorded the lowest revenue among ranked customers at **$9,129.35**.

Business Insight

The ranking analysis highlights a significant concentration of revenue among a small group of customers. The top-ranked customers contribute substantially more revenue than the majority of the customer base, emphasizing the importance of customer retention and relationship management.

Analyst Observation

Euro Shopping Channel generated approximately **$257,436** more revenue than Mini Gifts Distributors Ltd., the second-ranked customer. This demonstrates the exceptional value of the company's highest-performing customer and suggests that maintaining this relationship should be a strategic priority.

SQL Learning

This analysis used the `RANK()` window function to assign rankings based on revenue performance. Window functions are commonly used in business analytics for customer segmentation, performance tracking, and leaderboard reporting because they preserve individual records while performing advanced calculations across groups of data.

Best Sales Month in Each Year

This analysis identified the highest-performing sales month within each year using SQL window functions and partitioning.

Key Findings

- November 2003 generated **$1,029,837.66** in revenue and was the best-performing month of the year.
- November 2004 generated **$1,089,048.01** in revenue and was the highest-performing month in the entire dataset.
- May 2005 generated **$457,861.06** in revenue and ranked as the top-performing month for 2005.

Business Insight

The results reveal a strong seasonal sales pattern. November consistently emerged as the highest-performing month in both 2003 and 2004, suggesting that year-end demand plays a significant role in driving revenue.

Understanding these seasonal peaks can help the business improve inventory planning, marketing campaigns, and staffing decisions during high-demand periods.

Analyst Observation

The highest monthly revenue in the dataset was recorded in **November 2004**, generating over **$1.08 million** in sales. The repeated appearance of November as the strongest month indicates a recurring sales cycle rather than a one-time event.

SQL Learning

This analysis used the `PARTITION BY` clause within a window function. Partitioning allows SQL to create separate rankings within each year rather than ranking all months together, making it useful for year-over-year performance comparisons.

Top Products by Revenue

This analysis identified the individual products generating the highest revenue across all sales transactions.

Key Findings

- Product **S18_3232** (Classic Cars) generated the highest revenue at **$288,245.42**.
- Product **S10_1949** (Classic Cars) generated **$191,073.03** in revenue.
- Product **S10_4698** (Motorcycles) generated **$170,401.07**.
- Six of the top ten revenue-generating products belong to the **Classic Cars** product line.
- Product **S18_1662** from the Planes category generated **$139,421.97** in revenue.

Business Insight

The analysis confirms the dominance of the Classic Cars category, as the majority of the highest-performing products belong to this product line. This indicates strong customer demand and highlights the category as a major contributor to overall business performance.

Understanding which individual products drive revenue enables better inventory management, pricing decisions, and marketing strategies.

Analyst Observation

Product S18_3232 generated nearly **$288,000** in revenue, outperforming the second-ranked product by more than **$97,000**. The concentration of top-performing products within the Classic Cars category reinforces earlier findings that this product line is the primary revenue driver for the business.

SQL Learning

This analysis combined aggregation and sorting techniques to identify top-performing products. Product-level analysis is commonly used in business intelligence to support inventory planning, product development, and sales forecasting.

Order Status Analysis

This analysis examined order status categories to evaluate operational performance and identify potential risks within the sales process.

Key Findings

- Shipped orders accounted for **2,617 transactions** and generated **$9,291,501.08** in revenue.
- Cancelled orders accounted for **60 transactions** and generated **$194,487.48** in revenue.
- On Hold orders generated **$178,979.19** in revenue.
- Resolved orders generated **$150,718.28** in revenue.
- Disputed orders recorded the lowest revenue contribution at **$72,212.86**.

Business Insight

The overwhelming majority of revenue came from successfully shipped orders, indicating strong order fulfillment performance. More than **92% of total revenue** was generated through completed shipments.

However, cancelled, disputed, and on-hold orders collectively represent revenue that may be delayed, lost, or at risk. Monitoring these categories can help improve customer satisfaction and operational efficiency.

Analyst Observation

Shipped orders generated over **$9.29 million** in revenue, significantly outperforming all other order statuses combined. This suggests that the company's fulfillment process is operating effectively, although reducing cancellations and disputes could further improve revenue retention.

SQL Learning

This analysis used aggregation functions and grouping techniques to summarize business performance by operational status. Such analyses are commonly used to monitor fulfillment efficiency, customer service quality, and revenue leakage.

Conclusion

This project demonstrated the use of SQL for business-focused data analysis using a sales dataset containing over 2,800 transactions.

The analysis revealed that:

- The business generated over **$10 million** in total revenue.
- The United States was the highest-performing market, contributing more than **$3.6 million** in sales.
- Classic Cars emerged as the most profitable product line, generating nearly **$4 million** in revenue.
- November consistently represented the strongest sales period across multiple years.
- A small number of customers contributed a significant share of total revenue.
- Shipped orders accounted for the vast majority of business revenue.

Through this project, key SQL concepts including aggregation functions, filtering, grouping, ranking, window functions, partitioning, and query optimization were successfully applied to solve real-world business problems and generate actionable insights.

This analysis highlights the importance of SQL as a foundational tool for data analytics, business intelligence, and decision-making.

Query Optimization Concept

Although the dataset size was relatively small and did not require extensive optimization, indexing concepts were reviewed as part of the project requirements.

In larger production databases, indexes can be created on frequently queried columns to improve query performance and reduce execution time. Common candidates for indexing in this dataset would include CUSTOMERNAME, PRODUCTCODE, COUNTRY, STATUS, and ORDERDATE.

Understanding indexing is an important SQL skill because it helps improve database efficiency when working with large volumes of data.